

MAL Omics: Proteomics Analysis Report

COPDGene, SomaScan (~4,979 proteins), $n = 1,306$

Study. COPDGene is a longitudinal cohort study of current and former smokers enrolled at 21 US clinical centers. Subjects were seen at Phase 2 (~2013–2017) and Phase 3 (~2019–2022), approximately 5–6 years apart. Blood was drawn at Phase 2 and profiled by SomaScan, an aptamer-based platform that quantifies ~4,979 plasma proteins simultaneously as RFU (Relative Fluorescence Units). This analysis uses $n = 1,306$ subjects with complete proteomics, phenotype, and MAL data.

MAL (Mechanically Affected Lung). Computed tomography (CT)-derived continuous score (range 0.01–0.40) measuring the fraction of lung volume under destructive mechanical stress. Higher MAL = more lung tissue under stress. MAL is computed from inspiratory/expiratory CT image registration and is distinct from standard emphysema measures (e.g., %LAA-950). Variable name in data: `meanmal`.

MAL is itself a strong predictor of who will develop worse emphysema over time: higher mechanical stress on lung tissue drives progressive structural damage. However, MAL requires specialized CT analysis and is not available from a routine blood draw.

Goal. (1) Identify which Phase 2 plasma proteins are associated with MAL, establishing that mechanical lung stress leaves a detectable signal in blood. (2) Use those proteins to build a Proteomic Risk Score (proRS) from a Phase 2 blood draw that approximates MAL and can predict who will develop worse emphysema by Phase 3. If successful, the proRS would allow MAL-like risk stratification without requiring CT image registration. The proteins themselves are not measured at Phase 3; they serve as baseline predictors of the CT density change that occurs between visits (~6 years apart).

Outcome. Emphysema progression = change in CT lung density Phase 3 – Phase 2 (`Change_lung_density_vnb_P2_P3`, Hounsfield Units [HU]). Negative = worsened. Mean ≈ 1 HU (standard deviation [SD] 7) in this cohort; a negative value means the lung became less dense (more emphysematous) between visits.

Data Sources and Variable Names

Four CSV files were used. All joins are on `sid`.

`data/csv_exports/prot.csv`: SomaScan proteomics

Column	Description
<code>sid</code>	Subject identifier (join key)
<code>X{seqID}</code>	Raw RFU value for each protein (e.g. <code>X10000.28</code>). ~4,979 columns matching pattern <code>^X[0-9]</code> . Renamed to <code>log2_X{seqID}</code> after $\log_2$ transform.

data/csv_exports/pheno_MAL.csv: **Phenotypes and covariates**

Column	Used as
sid	Join key
Change_lung_density_vnb_P2_P3	Outcome: emphysema progression (HU)
lung_density_vnb_P2	Baseline lung density (HU)
Age_P2	Age (years)
gender	Sex (coded 1/2)
race	Race (coded 1/2)
ATS_PackYears_P2	Pack-years of smoking
SmokCigNow_P2	Current smoking (0/1)
scanner_model_clean_P2	CT scanner model
FEV1_FVC_post_P2	FEV ₁ (forced expiratory volume in 1 second) / FVC (forced vital capacity)
PC1 – PC5	Ancestry principal components
fev1_pp_gli_global_P2	FEV ₁ % predicted (GLI global reference)
BMI_P2	Body mass index

data/Data/MAL.csv: **MAL scores**

Column	Description
sid	Subject identifier (join key)
meanmal	MAL score (continuous, range 0.01–0.40)

data/csv_exports/CellData.csv: **Phase 2 BMI**

Column	Used as	Note
sid	Join key	
Phase_study	Phase filter	Rows with Phase_study == 2 used
BMI	BMI_P2 in analysis	All 1,306 analysis subjects present, no missingness

data/csv_exports/metadat5k.csv: **Protein annotation**

Column	Description
seqID	Protein sequence identifier; joins to log2_X{seqID}
EntrezGeneSymbol	HGNC (HUGO Gene Nomenclature Committee) gene symbol (e.g. ADIPOQ)
Target	Short protein name (e.g. Adiponectin)
TargetFullName	Full protein name

Join logic. Three-way inner join: `prot ⋈sid pheno ⋈sid MAL`. Metadata joined on `seqID = sub("^log2.X", "", predictor)`.

File sizes: `prot.csv` has 5,670 subjects; `MAL.csv` has 5,024 subjects; `pheno_MAL.csv` has 10,652 subjects. After the 3-way inner join on `sid`: $n = 4,630$.

Complete-case filter then requires non-missing values for all of: `Change_lung_density_vnb_P2_P3` (Phase 3 outcome), `lung_density_vnb_P2`, `meanmal`, `Age_P2`, `gender`, `race`, `ATS_PackYears_P2`,

SmokCigNow_P2, scanner_model_clean_P2, fev1_pp_gli_global_P2, FEV1_FVC_post_P2, and PC1-PC5.

This reduces the sample to $n = 1,306$. The large drop ($4,630 \rightarrow 1,306$) is driven primarily by the Phase 3 outcome: 8,912 of 10,652 subjects in `pheno_MAL.csv` (84%) have no Phase 3 CT scan recorded, meaning they did not return for the follow-up visit ~ 6 years later.

Table 1: Demographics

Study population after 3-way merge (proteomics $\times$ phenotype $\times$ MAL) and complete-case filtering.

Table 1. Baseline characteristics of COPDGene participants included in the proteomics analysis ($n = 1,306$). Subjects had complete SomaScan plasma proteomics, phenotype data, and MAL score. Continuous variables: mean (SD); categorical: n (%). MAL (Mechanically Affected Lung) = CT-derived score (range 0.01–0.40); higher values indicate more lung under destructive mechanical stress. Emphysema progression = CT lung density at Phase 3 minus Phase 2 ($\rho_{P3} - \rho_{P2}$, HU); negative values indicate worsening.

Characteristic	Overall ($n = 1,306$)
<i>Demographics</i>	
Age (years)	65 (8)
Sex: male / female	633 (48%) / 673 (52%)
Race: White / Black	991 (76%) / 315 (24%)
<i>Smoking</i>	
Current smoker	445 (34%)
Pack-years	43 (23)
<i>Pulmonary function</i>	
FEV ₁ % predicted (GLI global)	83 (24)
FEV ₁ /FVC	0.69 (0.14)
<i>CT imaging</i>	
Lung density Phase 2 (HU)	83 (23)
MAL score	0.04 (0.05)
Emphysema progression (HU)	1 (7)
<i>Anthropometrics</i>	
BMI (kg/m ²)	29 (6)

Note. Sex is coded 1/2 in the data; assumed Male/Female per COPDGene convention. Race coded 1/2; assumed White/Black. Emphysema progression mean of 1 HU (SD 7) reflects a near-zero mean; the distribution includes both progressors (negative $\Delta\rho$) and improvers (positive $\Delta\rho$).

Figure 1: Study Design

Requires a schematic (e.g., BioRender) illustrating the analysis flow: SomaScan proteomics $\rightarrow$ differential expression $\rightarrow$ GSEA (Gene Set Enrichment Analysis) / STRING $\rightarrow$ adaptive LASSO

(Least Absolute Shrinkage and Selection Operator) proRS (Proteomic Risk Score) → emphysema progression association → causal mediation.

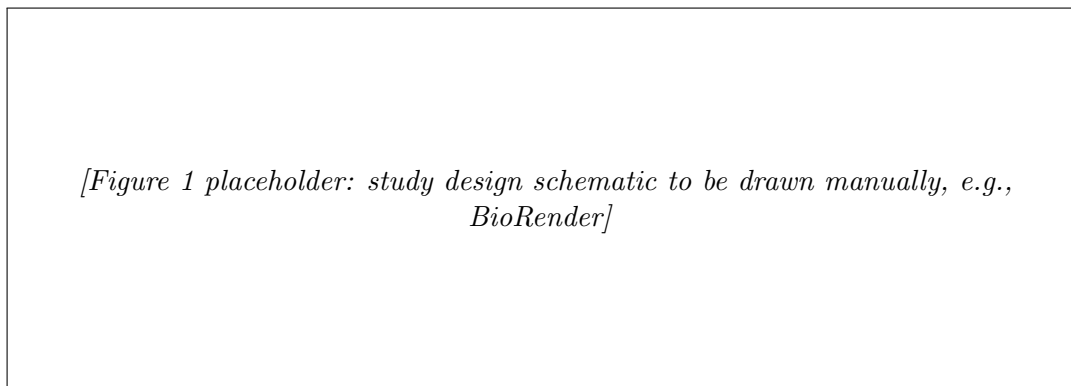


Figure 1: Overview of the COPDGene MAL omics analysis pipeline. SomaScan plasma proteomics ($\sim 4,979$ proteins, Phase 2) are used to identify proteins associated with MAL (differential expression, GSEA, STRING network), build a proteomic risk score (adaptive LASSO), test whether the score predicts emphysema progression (Phase 3 minus Phase 2 CT density), and quantify mediation of the MAL effect through blood proteins (causal mediation analysis).

Figure 2: Protein Expression Landscape of MAL

A. Volcano Plot (protein $\sim$ MAL)

Each dot is one protein. The x -axis ($\hat{\beta}$) shows how much a protein changes per one-unit increase in MAL after adjusting for age, sex, race, smoking, pack-years, BMI, scanner, and ancestry PCs. Positive $\hat{\beta}$ = protein is higher with more mechanical lung stress; negative = lower. $\hat{\beta}$ is defined as the change in $\log_2$ protein per *one-unit* increase in MAL. But MAL only ranges from 0.01 to 0.40 in this cohort, so a one-unit change never actually occurs. To express the effect in realistic terms, multiplying $\hat{\beta}$ by the full MAL range ($0.40 - 0.01 = 0.39$) gives the maximum expected protein change across the entire observed spread of MAL. For TNNT2 (Troponin T, the most significant protein): $0.227 \times 0.39 \approx 0.09 \log_2$ -unit shift from the lowest to the highest MAL subject. The y -axis ($-\log_{10} p$ -value) is confidence: higher = more significant. The dashed horizontal line marks the $FDR < 0.05$ threshold on the p -value scale. Dots above the threshold are coloured red (enriched) or blue (depleted); grey dots did not reach significance. A dot far right and high = reliably elevated with more MAL. A dot far left and high = reliably lower with more MAL.

How calculated. For each of 4,979 proteins:

$$\log_2(\text{protein}) \sim \beta_{\text{MAL}} \cdot \text{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{BMI} + \text{scanner} + \text{PC1} - \text{PC5}$$

$\hat{\beta}_{\text{MAL}}$ = change in $\log_2$ protein level per one-unit increase in MAL. FDR correction (Benjamini–Hochberg) across all 4,979 tests. $\text{FEV}_1\%$ predicted is intentionally excluded: airflow obstruction lies on the causal pathway from mechanical stress to lung damage, so adjusting for it would remove the signal of interest.

Key result. 14 of 4,979 proteins significant at $FDR < 0.05$ (6 positively associated, 8 negatively associated).

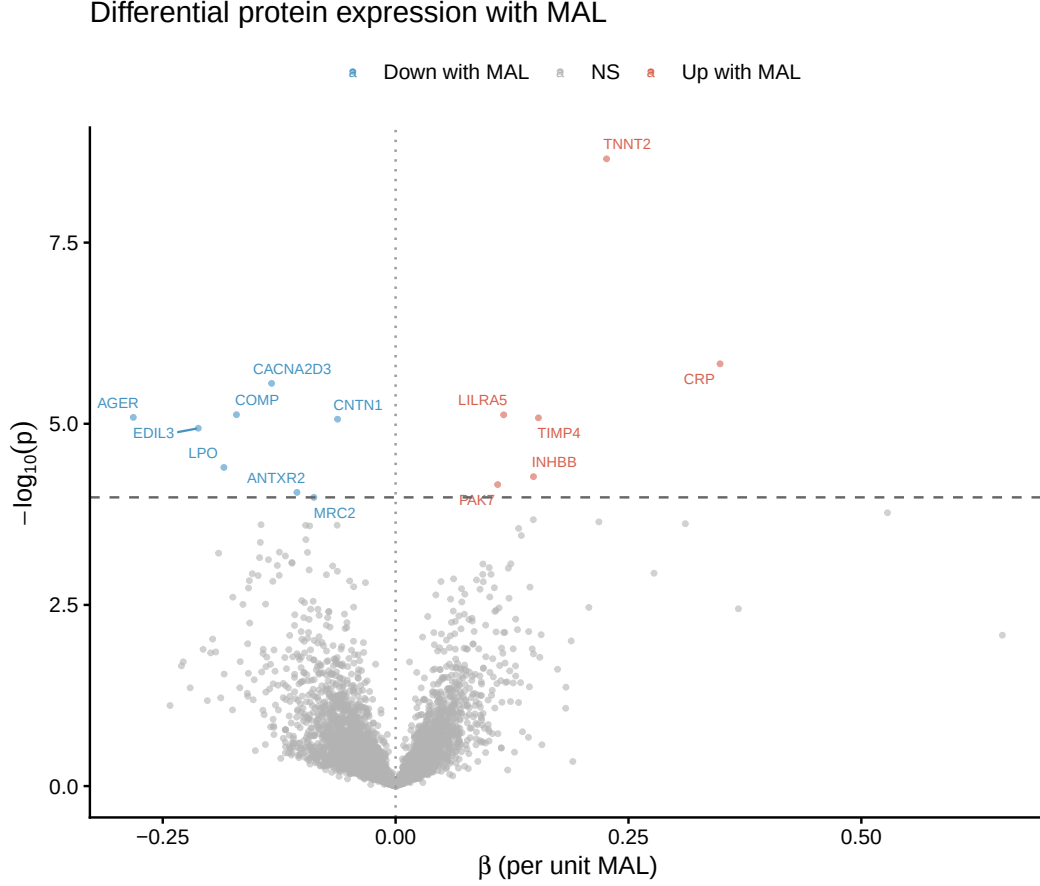


Figure 2: Differential plasma protein expression associated with Mechanically Affected Lung (MAL) score in COPDGene participants ($n = 1,306$; 4,979 SomaScan proteins). Each point is one protein. x -axis: regression coefficient ($\hat{\beta}$) for $\log_2$ protein level per unit MAL, adjusted for age, sex, race, pack-years, smoking, BMI, CT scanner, and ancestry PCs (PC1–PC5). y -axis: $-\log_{10}(p\text{-value})$; dashed line = FDR < 0.05 threshold. Red = FDR < 0.05, positively associated; blue = FDR < 0.05, negatively associated; grey = not significant. Top 20 proteins by FDR labelled. 14 proteins reached FDR < 0.05 (6 positively, 8 negatively associated with MAL).

Top enriched (higher with more MAL, FDR < 0.05): *TNNT2* (Troponin T, $\beta = 0.23$), *CRP* ($\beta = 0.35$), *TIMP4* ($\beta = 0.15$), *LILRA5* ($\beta = 0.12$), *INHBB* ($\beta = 0.15$), *PAK7* ($\beta = 0.11$).

Top depleted (lower with more MAL, FDR < 0.05, ordered by FDR): *CACNA2D3* ($\beta = -0.13$), *CNTN1* (contactin-1, $\beta = -0.06$), *AGER* (sRAGE, $\beta = -0.28$), *COMP* ($\beta = -0.17$), *EDIL3* ($\beta = -0.21$), *LPO* ($\beta = -0.18$), *ANTXR2* ($\beta = -0.11$), *MRC2* ($\beta = -0.09$).

A2. Sensitivity Analysis: Adjusting for CAD and CHF

Several MAL-associated proteins (e.g., *TNNT2*, *CRP*) are established cardiovascular biomarkers, raising the question of whether the observed signals reflect cardiovascular comorbidity rather than mechanical lung injury. To evaluate this, the primary differential expression analysis was repeated with two additional covariates: history of coronary artery disease (CAD; *CoronaryArtery_P2*) and history of congestive heart failure (CHF; *congestheartfail_slv_P2*), both ascertained at the

Phase 2 visit from `CellData.csv`. CAD and CHF data were complete for all $n = 1,306$ analysis subjects (CAD: 7.5%, $n = 98$; CHF: 1.1%, $n = 15$).

The model for each protein was:

$$\log_2(\text{protein}) \sim \text{MAL} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{BMI} + \text{scanner} + \text{PC1-PC5} + \text{CAD} + \text{CHF}$$

Results are saved to `mal_de_sensitivity_cad_chf.csv`.

Result. All 14 proteins significant in the primary analysis remained significant at $\text{FDR} < 0.05$ after CAD/CHF adjustment (100% overlap; 0 proteins lost). Regression coefficients were virtually unchanged (median absolute $\Delta\hat{\beta} = 1.1\%$; $r = 1.00$ between primary and adjusted estimates), indicating that MAL-associated protein signals are not explained by prevalent cardiovascular disease.

B. GSEA Barplot

GSEA asks whether an entire biological pathway shows a coordinated shift with MAL, rather than asking protein by protein. Each bar is one significant pathway. Bar pointing right ($\text{NES} > 0$): the pathway's proteins are on average higher in plasma with higher MAL. Bar pointing left ($\text{NES} < 0$): lower. Longer bar = stronger shift. NES of ± 2 is a strong, consistent signal. padj is the Benjamini-Hochberg adjusted p -value across all pathways tested; only bars with $\text{padj} < 0.05$ are shown.

How calculated. All 4,979 proteins ranked by their MAL t -statistic (most positively associated at top, most negatively at bottom). Gene sets tested: Hallmark, KEGG Legacy, Reactome (gene-set sizes 15–500).

NES (Normalized Enrichment Score): measures whether a gene set's members concentrate at the top or bottom of the ranked list. $\text{NES} > 0$ = enriched in high-MAL subjects; $\text{NES} < 0$ = depleted. Normalization accounts for gene-set size so scores are comparable across pathways.

padj (adjusted p -value): `fgsea` tests hundreds of pathways at once. Without correction, many would appear significant by chance. padj is the Benjamini-Hochberg correction applied within each gene set collection (Hallmark, KEGG, Reactome separately). A $\text{padj} < 0.05$ means fewer than 5% of pathways at this threshold are expected to be false positives. Only pathways with $\text{padj} < 0.05$ are shown in the barplot.

Results. 11 pathways significant ($\text{padj} < 0.05$). Eight are depleted ($\text{NES} < 0$): glycosaminoglycan/keratan sulfate metabolism, carbohydrate and lipid metabolism, biological oxidations, branched-chain amino acid degradation, and fatty acid metabolism. Three are enriched ($\text{NES} > 0$): antimicrobial peptides, chemokine receptor signalling, and complement/coagulation cascades (inflammatory/immune). Higher MAL is associated with lower circulating metabolic proteins and higher inflammatory proteins.

Gene Set Enrichment Analysis (GSEA): significant pathways associated with MAL. Proteins ranked by MAL t -statistic ($n = 4,979$) and tested using `fgsea` against Hallmark, KEGG Legacy, and Reactome gene sets (sizes 15–500). NES = Normalized Enrichment Score; NES > 0 = pathway proteins higher in high-MAL subjects (enriched); NES < 0 = lower (depleted). padj = Benjamini–Hochberg adjusted p -value within each collection. Only pathways with padj < 0.05 shown ($n = 11$).

Collection	Pathway	NES	padj
Reactome	Diseases Assoc. with Glycosaminoglycan Metabolism	−2.07	0.011
Reactome	Keratan Sulfate / Keratin Metabolism	−2.02	0.022
KEGG	Valine, Leucine and Isoleucine Degradation	−2.03	0.028
Reactome	Biological Oxidations	−1.85	0.022
Reactome	Sensory Perception	−1.86	0.022
Reactome	Diseases of Metabolism	−1.87	0.011
Reactome	Metabolism of Carbohydrates and Derivatives	−1.76	0.019
Hallmark	Fatty Acid Metabolism	−1.69	0.039
Reactome	Antimicrobial Peptides	+2.06	0.011
Reactome	Chemokine Receptors Bind Chemokines	+2.19	0.011
KEGG	Complement and Coagulation Cascades	+1.89	0.028

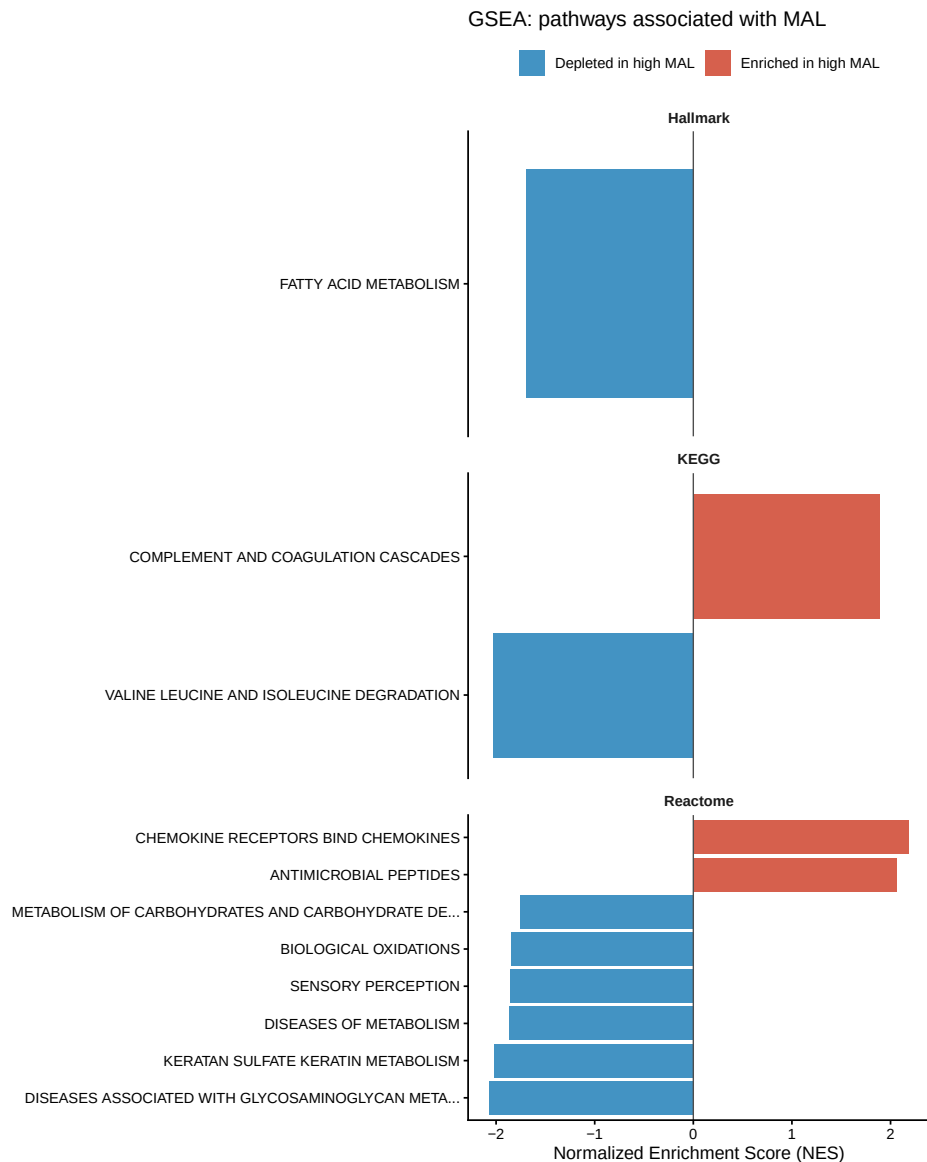


Figure 3: Gene Set Enrichment Analysis (GSEA) of MAL-associated plasma proteins across Hallmark, KEGG Legacy, and Reactome collections. Proteins ranked by MAL t -statistic ($n = 4,979$); bar length represents the Normalized Enrichment Score (NES). NES > 0 = enriched in high-MAL subjects; NES < 0 = depleted. Colour indicates direction. Only pathways with Benjamini–Hochberg adjusted $p < 0.05$ are shown ($n = 11$; 8 depleted, 3 enriched).

C. STRING Network

Protein–protein interaction network for the 14 MAL-associated proteins. Each node is one protein; node colour indicates direction of association with MAL (red = enriched, blue = depleted) and node size encodes $-\log_{10}(\text{FDR})$. An edge between two proteins indicates a known or predicted functional interaction in STRING. Edge thickness scales with the STRING combined confidence score. A cluster of connected nodes indicates those proteins share a biological process; isolated nodes had no qualifying interactions within this set.

How calculated. 14 FDR-significant proteins submitted to STRING v11.5 human database (human, species 9606). All interactions with combined confidence score ≥ 200 drawn to reveal subnetwork structure; edge thickness proportional to score. Layout: Kamada-Kawai algorithm (`igraph::layout_with_kk`).

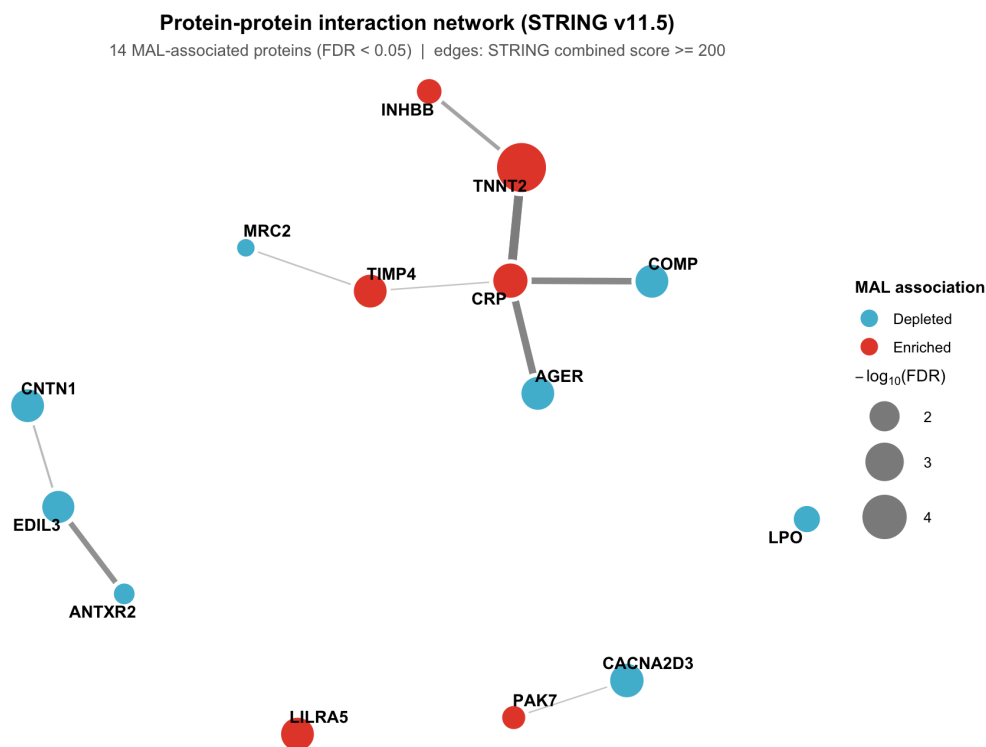


Figure 4: Protein-protein interaction network for the 14 plasma proteins significantly associated with MAL (FDR < 0.05), constructed using STRING v11.5 (human, species 9606). Node colour: red = enriched with MAL, blue = depleted. Node size: $-\log_{10}(\text{FDR})$. Edges: STRING combined confidence score ≥ 200 ; thickness proportional to score. Layout: Kamada-Kawai.

C2. MCL Clustering and Cluster Enrichment

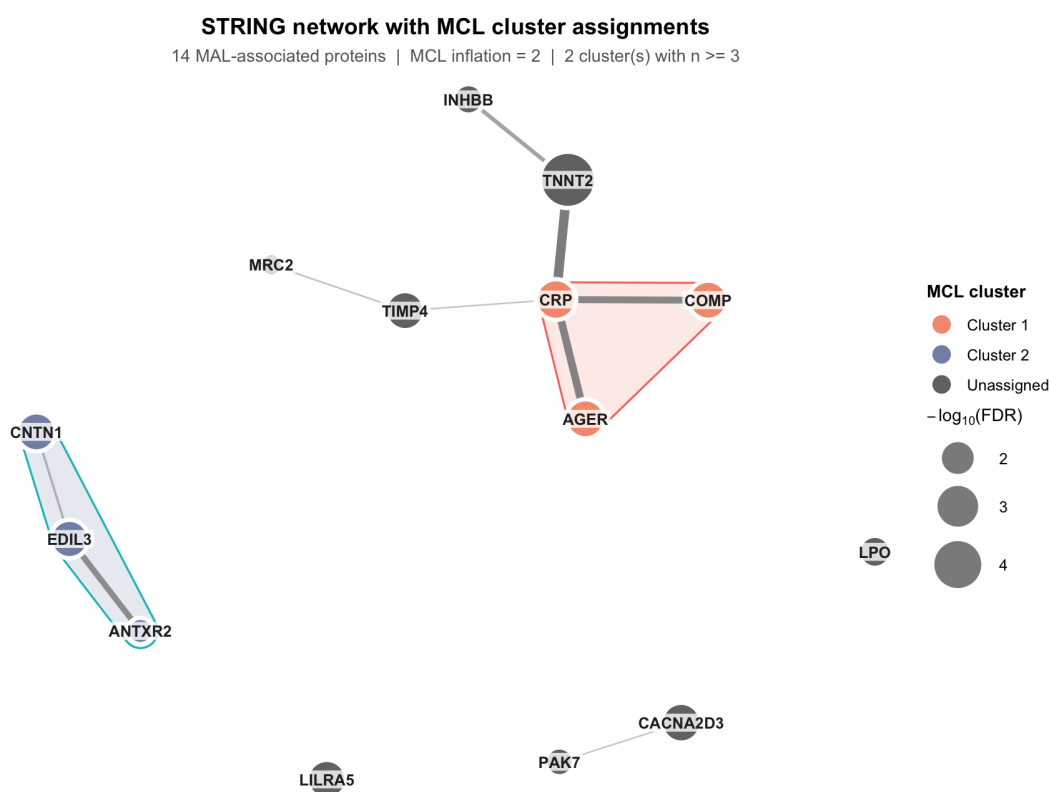
Markov Cluster Algorithm (MCL) clustering (inflation parameter = 2, default) was applied to the STRING interaction network of the 14 MAL-associated proteins. MCL was run using the MCL R package (`mcl()`, `addLoops=TRUE`, `inflation=2`) on the binary adjacency matrix of the simplified subnetwork (combined score ≥ 200). Only clusters with ≥ 3 proteins were retained; pathway enrichment was retrieved from the STRING API across Gene Ontology, KEGG, Reactome, and PubMed literature co-citation (text mining); FDR computed by STRING using Benjamini-Hochberg against the human proteome background. Full results are in `STRING_mcl_clusters.csv` and `STRING_cluster_enrichment.csv`.

Two clusters with ≥ 3 proteins emerged. Cluster 1 ($n = 3$: AGER, COMP, CRP) co-enriches for published biomarker panels of carotid stenosis and osteoarthritis (FDR = 0.006), reflecting shared involvement in vascular and cartilage stress signalling. Cluster 2 ($n = 3$: ANTXR2, CNTN1, EDIL3) comprises extracellular matrix and cell adhesion proteins; no significant pathway enrich-

ment was returned by STRING for this cluster. The remaining 8 proteins were unassigned.

MCL cluster enrichment summary. Top enriched term per cluster (lowest FDR). Clusters with ≥ 3 proteins shown. Full results in `STRING_cluster_enrichment.csv`.

Cluster	<i>n</i>	Members	Top term (source)	FDR
1	3	AGER, COMP, CRP	Biomarker panel for carotid stenosis / osteoarthritis (PubMed co-citation)	0.006
2	3	ANTXR2, CNTN1, EDIL3	No significant enrichment	—



STRING network with MCL cluster assignments. Nodes represent the 14 MAL-associated proteins ($\text{FDR} < 0.05$); edges: STRING combined score ≥ 200 ; thickness proportional to score. Shaded hull indicates MCL cluster membership (inflation = 2). Cluster 1 (orange, $n = 3$): AGER, COMP, CRP; co-cited as vascular/ cartilage biomarkers in PubMed ($\text{FDR} = 0.006$). Cluster 2 (purple, $n = 3$): ANTXR2, CNTN1, EDIL3; extracellular matrix and cell adhesion proteins; no significant pathway enrichment. Remaining 8 proteins unassigned (dark grey). Layout: Kamada–Kawai.

Proteomic Risk Score: Methods

A single proteomic risk score is constructed for mechanical lung stress and used for both the AIC model comparison and the causal mediation analysis.

MAL proteomic risk score (proRS). An adaptive LASSO ($\alpha = 1$) with MAL score (`meanmal`) as the outcome was fitted on 70% of the cohort (`set.seed(42)`, $n_{\text{train}} = 914$, $n_{\text{test}} = 392$). SomaScan RFU values were $\log_2$ -transformed. Clinical covariates (age, sex, race, pack-years, smoking status, CT scanner type, PC1–PC5, BMI) were included as unpenalised fixed effects. Ridge regression on residualised `meanmal` produced adaptive weights $w_j = 1/|\hat{\beta}_j^{\text{ridge}}|$; 10-fold inner cross-validation selected $\lambda_{\min}$; 572 proteins were retained. Test $R^2 = 0.233$.

Post-selection OLS refitting yielded unbiased $\hat{\beta}_j$, SE, 95% CI, and p -values for each protein. The standardised score is:

$$\text{proRS}_i = \frac{\sum_j \hat{\beta}_j \cdot \log_2(\text{protein}_{ij}) - \bar{x}}{s}$$

where $\bar{x}$ and s are the mean and SD computed across all $n = 1,306$ subjects (`MAL_PRS_scaled`). The cross-validation curve (Supplementary Figure C) shows MSE as a function of $\log(\lambda)$, and Supplementary Figure D shows the concordance between `MAL_PRS_scaled` and CT-derived `meanmal` ($R^2 = 0.543$, $n = 1,306$).

Association of Proteomic Risk Score with Emphysema Progression

The MAL proRS (`MAL_PRS_scaled`; 572 proteins, test $R^2 = 0.233$ vs. `meanmal`) is evaluated on the held-out 30% test set ($n = 392$) to test whether the mechanistic protein signature of MAL independently predicts future emphysema progression beyond clinical risk factors. Three nested linear regression models use `pctprog` as the outcome; scanner is included in M2 as a technical confounder.

M1 (Clinical): `pctprog` $\sim$ age + sex + race + smoking + pack-years + BMI + scanner

M2 (proRS + scanner): `pctprog` $\sim$ proRS + scanner

M3 (Clinical + proRS): `pctprog` $\sim$ proRS + age + sex + race + smoking + pack-years + BMI + scanner

Model comparison for predicting emphysema progression (`pctprog`) by AIC (test set, $n = 392$; lower = better fit). Proteomic Risk Score = adaptive LASSO targeting `meanmal` (`MAL_PRS_scaled`, $\alpha = 1$, 572 proteins, 10-fold inner CV; trained on 70%, evaluated on 30%). ΔAIC relative to M1 (clinical reference); $\Delta \leq -2$ indicates meaningful improvement.

Model	AIC	ΔAIC
M1: Clinical (age, sex, race, smoking, pack-years, BMI, scanner)	2,259.2	Reference
M2: proRS + scanner	2,286.6	+27.4
M3: Clinical + proRS	2,245.7	-13.5

Effect sizes for `MAL_PRS_scaled`:

- M2: $\hat{\beta} = 1.09\%$, 95% CI [0.67, 1.51], $p < 0.001$
- M3: $\hat{\beta} = 0.81\%$, 95% CI [0.40, 1.21], $p < 0.001$

Key result. The MAL proteomic risk score adds independent predictive value for emphysema progression beyond clinical risk factors (M3: $\hat{\beta} = 0.81\%/SD$, $p < 0.001$; $\Delta AIC = -13.5$, exceeding the ≤ -2 threshold). These proteins were selected to predict CT-derived mechanical lung stress, not emphysema progression, yet the score independently predicts future CT density loss beyond smoking history, age, BMI, and spirometry. This indicates that the blood protein signature of mechanical lung stress carries prognostic information not captured by standard clinical covariates.

Figure 3: Proteomic Signature of Mechanical Stress Predicts Emphysema Progression

Figure 3 shows a dose-response analysis of the MAL proteomic risk score against emphysema progression. Subjects were divided into 10 equal deciles of `MAL_PRS_scaled` (≈ 131 subjects per decile; D1 = lowest score, D10 = highest). The x-axis shows the mean proteomic risk score within each decile bin (SD units); the y-axis shows the mean `pctprog` (percentage of lung voxels showing emphysema progression) for that bin. A linear regression line with 95% confidence band is overlaid. Error bars on each point are 95% CI (mean $\pm 1.96 \times SE$). No covariates were adjusted.

How calculated. `MAL_PRS_scaled` is the standardised adaptive LASSO score described in the Proteomic Risk Score Methods section (572 proteins, outcome = `meanmal`, trained on 70% of the cohort). All $n = 1,306$ subjects are ranked by their score and divided into 10 equal-sized groups (deciles; ≈ 131 subjects each). Within each decile, the mean score (x) and mean `pctprog` (y) are computed. Error bars are 95% CI (mean $\pm 1.96 \times SE$, where $SE = SD/\sqrt{n}$). The overlaid line is a simple linear regression of `pctprog` on `MAL_PRS_scaled` across all $n = 1,306$ subjects; the shaded band is the 95% confidence region. No covariates are adjusted in this descriptive plot. The D10/D1 ratio compares the mean `pctprog` in the highest vs. lowest decile; the trend p -value is from the regression slope.

Key result. The proteomic risk score shows a near-monotone dose-response with emphysema progression: subjects in the highest decile of the blood score had $3.1\times$ more lung progression than those in the lowest (D10/D1 = $3.1\times$, $p = 1.52 \times 10^{-40}$). This dose-response arises from a score derived entirely from plasma proteins targeting CT-measured mechanical stress, demonstrating that the blood signature of MAL independently stratifies patients by future emphysema progression.

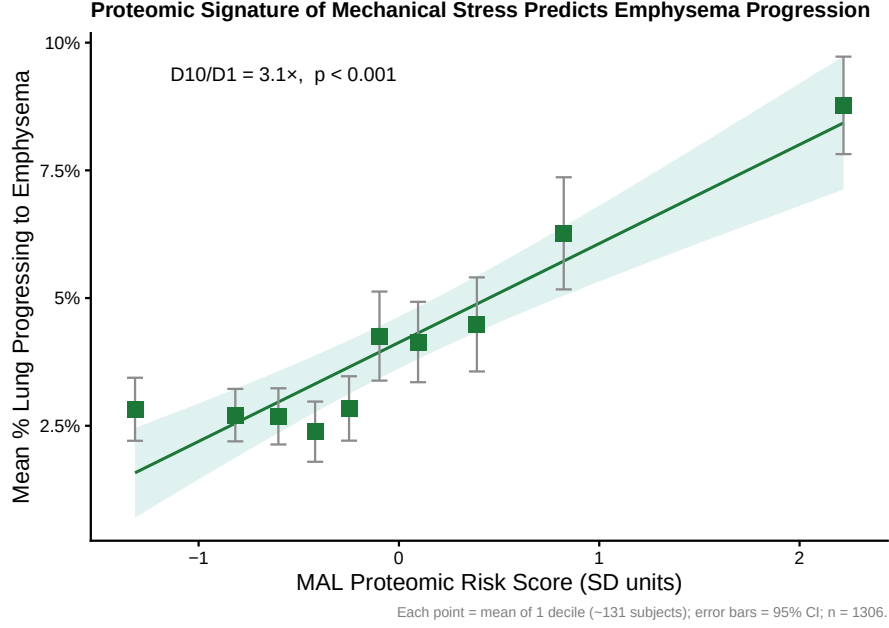


Figure 5: Mean % of lung progressing to emphysema (`pctprog`, y-axis) across 10 equal deciles of the MAL proteomic risk score (`MAL_PRS_scaled`, x-axis, SD units; $n = 1,306$). Each green square is the mean of one decile (≈ 131 subjects); error bars: 95% CI of the decile mean. Line: linear regression of `pctprog` on `MAL_PRS_scaled` across all subjects; shaded band: 95% confidence region for the regression line (i.e., uncertainty in the fitted mean at each score value, not a prediction interval for individual subjects). $D10/D1 = 3.1\times$, $p = 1.52 \times 10^{-40}$.

Table 2: Causal Mediation

The analysis decomposes the total effect of MAL on emphysema progression into a direct path (NDE) and a path through the blood protein score (NIE). NDE = how much of the MAL effect goes through pathways other than the protein score. NIE = how much goes through the protein score. All effects are in HU (Hounsfield Units), the CT density scale. More negative HU = lower lung density = more emphysema. A difference of 1 HU is a small change; the SD of progression in this cohort is ~ 7 HU.

- **NDE = -0.55 HU**, $p = 0.941$: MAL’s direct effect on progression is non-significant once blood protein changes are accounted for. The 95% CI spans zero $[-14.96, 13.86]$, consistent with no independent direct effect.
- **NIE = 13.93 HU**, $p = 0.002$: the protein risk score mediates a large and significant portion of the MAL effect. A one-unit increase in MAL is associated with 13.93 HU more progression through the protein pathway.
- **Proportion mediated $\approx 100\%$** : the indirect path through blood proteins explains essentially all of the MAL–progression relationship. The NDE is non-significant, consistent with complete mediation.
- **Total effect = 13.38 HU**: NDE + NIE; the protein pathway accounts for the entire effect.

How calculated. Pathway tested: MAL $\rightarrow$ Proteomic Risk Score $\rightarrow$ Emphysema Progression.

Fitted using the `medflex` R package (natural effects framework) with robust sandwich standard errors.

Two models fitted internally by `medflex`:

Imputation model: $\Delta\rho \sim \text{meanmal} \times \text{proRS} + \text{covariates}$

Natural effects model: $\Delta\rho \sim \text{meanmal}_0 + \text{meanmal}_1 + \text{covariates}$

`meanmal0` is the MAL value at which the mediator (proRS) is set (direct path); `meanmal1` is the actual MAL value driving the outcome (indirect path). The difference between them isolates the mediated portion.

Table 2. Causal mediation of the MAL–emphysema progression relationship through the Proteomic Risk Score (proRS), estimated using the natural effects framework (`medflex` R package, robust sandwich SE, $n = 1,306$). NDE (Natural Direct Effect) = portion of MAL’s effect on ΔCT lung density not operating through the proRS. NIE (Natural Indirect Effect) = portion operating through the proRS. Proportion mediated = NIE / total effect. All effects in HU (Hounsfield Units; negative = more emphysema).

Effect	Estimate (HU)	95% CI	z	p
Natural Direct Effect (NDE)	−0.55	[−14.96, 13.86]	−0.07	0.941
Natural Indirect Effect (NIE)	+13.93	[5.26, 22.60]	+3.15	0.002
Total effect	+13.38	—		
Proportion mediated		≈100%		

Key result. The proteomic risk score fully mediates the effect of MAL on emphysema progression (NIE = 13.93 HU, $p = 0.002$). The direct path is non-significant once the protein pathway is accounted for (NDE = −0.55 HU, $p = 0.941$), consistent with complete mediation.

Supplement

Supplementary Tables

- **S1: All differential expression results** (4,979 proteins): gene, target, $\hat{\beta}$, SE, t , p , FDR.
File: MAL_omics/mal_de_results.csv
- **S2: proRS weights** (MAL-targeting LASSO proteins): gene, target, $\hat{\beta}_{\text{OLS}}$, 95% CI, p .
File: MAL_omics/mal_risk_score_weights.csv
- **S3: Per-protein mediation** (MAL-targeting LASSO proteins): NDE, NIE, total, proportion mediated, NIE p .
File: MAL_omics/mal_mediation_supp.csv

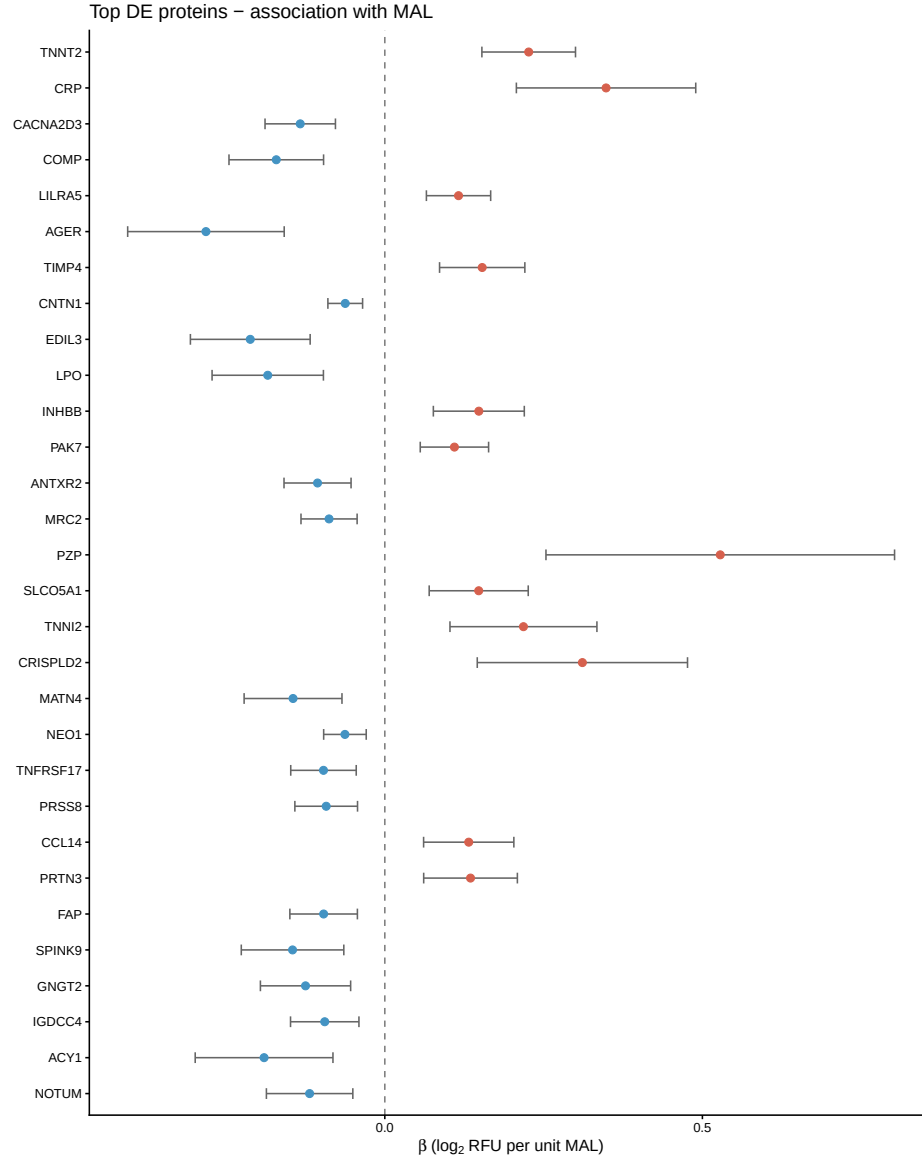
Of 199 converged proteins (from the 572 proRS proteins), 10 reached NIE $p < 0.05$. Proportions mediated range from −9% to 18%; EPHB2 shows negative mediation (suppression).

Supplementary Table S3. Per-protein causal mediation: MAL $\rightarrow$ individual protein $\rightarrow$ emphysema progression. Only proteins with NIE $p < 0.05$ shown ($n = 10$ of 202 converged). NDE = Natural Direct Effect; NIE = Natural Indirect Effect. All effects in HU. Proportion mediated = NIE / (NDE + NIE). `medflex`, robust SE, $n = 1,306$.

Gene	Target	NDE (HU)	NDE p	NIE (HU)	NIE p	% Mediated
TREM2	TREM2	+13.67	0.011	+2.78	0.012	17%
MANSC4	MANS4	+14.23	0.008	+1.73	0.012	11%
EPHB2	EPHB2	+20.54	< 0.001	-1.64	0.016	-9%
DEFA5	HD-5	+14.61	0.006	+1.66	0.017	10%
ADH7	ADH7	+15.01	0.006	+1.79	0.021	11%
ADAMTS5	ADAMTS-5	+15.37	0.004	+1.22	0.021	7%
INHBB	Inhibin β B	+13.07	0.020	+2.88	0.022	18%
SCG3	SCG3	+13.46	0.015	+2.65	0.029	16%
MSMP	PSMP	+13.80	0.011	+1.78	0.034	11%
PDCD1	PD-1	+16.26	0.002	+0.76	0.048	4%

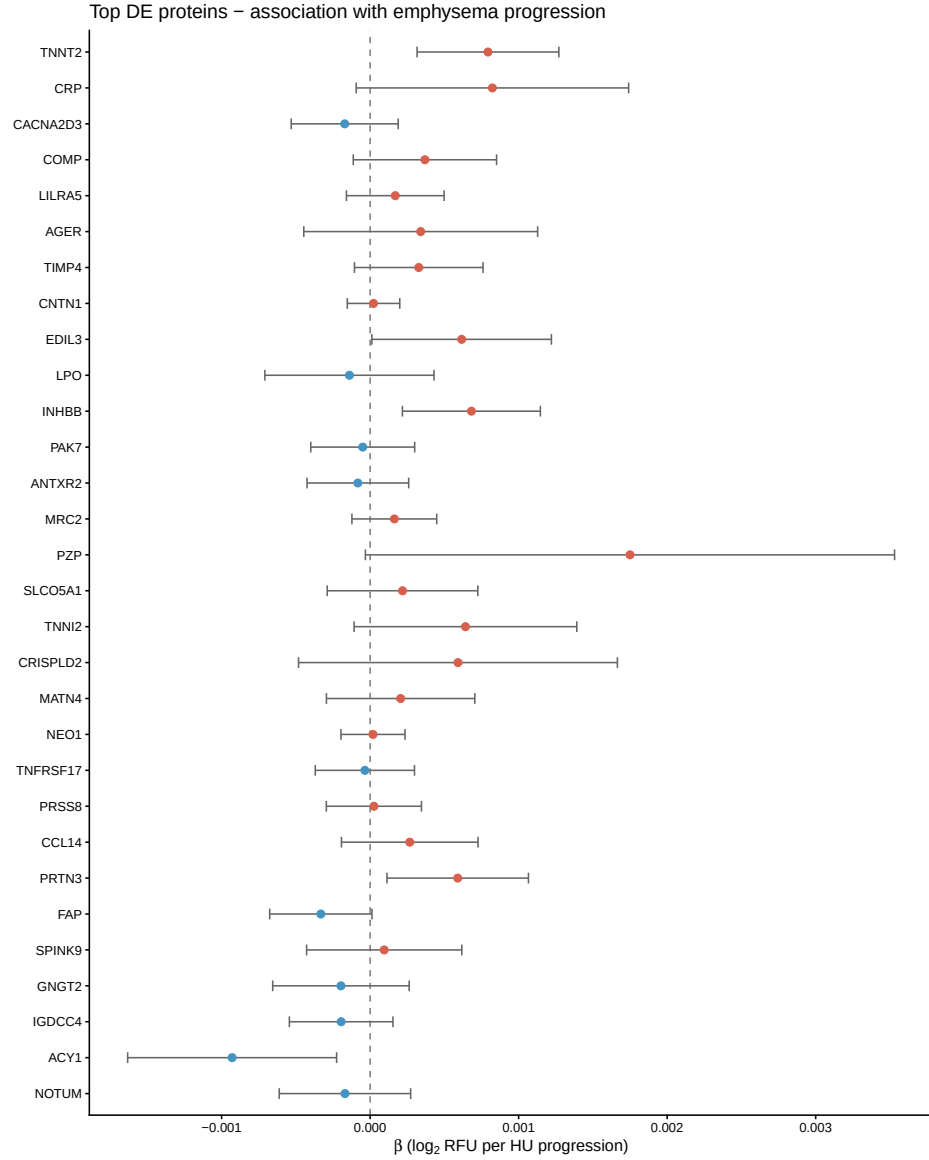
Supplementary Figures

- **Supp Figure A: Forest plot, proteins $\sim$ MAL.** Top 30 proteins by $|t\text{-statistic}|$, showing $\hat{\beta}$ ($\log_2$ RFU per unit MAL) with 95% CI. Red = positively associated; blue = negatively associated with MAL.



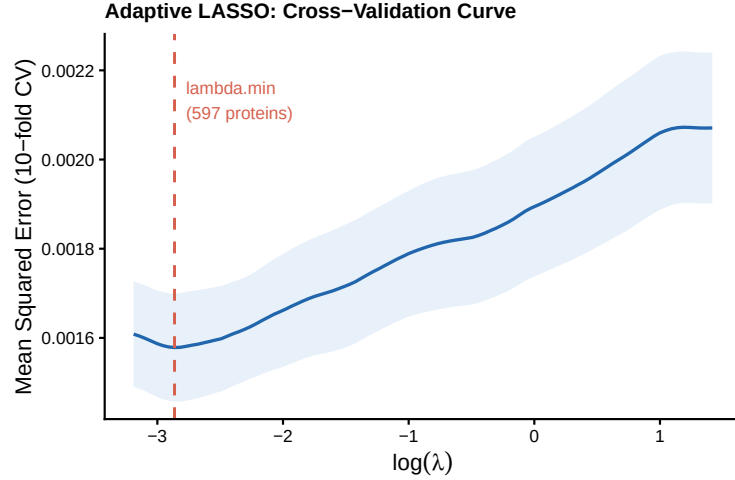
Supplementary Figure A. Forest plot of the top 30 proteins most significantly associated with MAL score (ranked by $|t\text{-statistic}|$). Points show $\hat{\beta}$ (log₂ RFU per unit MAL) with 95% CI, from the model: $\log_2(\text{protein}) \sim \text{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{BMI} + \text{scanner} + \text{PC1} - \text{PC5}$. Red = positively associated; blue = negatively associated with MAL. Vertical dashed line at zero.

- **Supp Figure B: Forest plot, proteins ~ emphysema progression.** Same 30 proteins, association with emphysema progression ($\text{Change_lung_density_vnb_P2_P3}$) adjusted for baseline lung density and covariates.



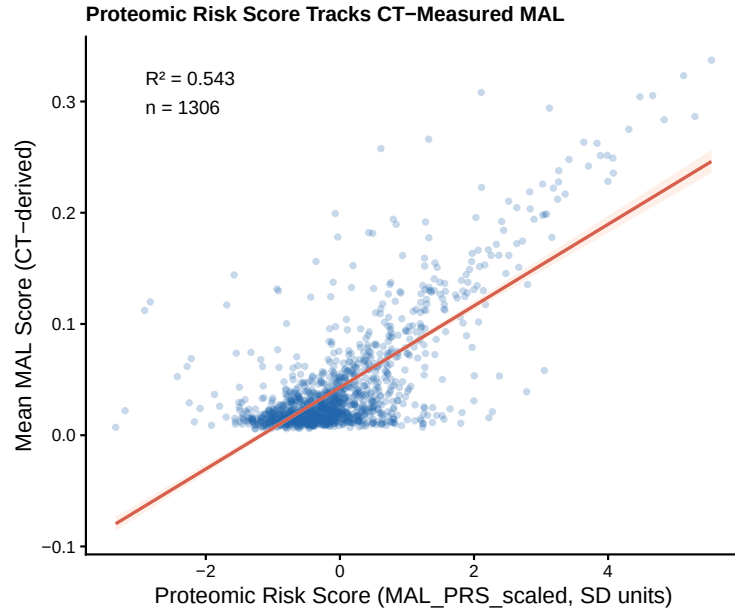
Supplementary Figure B. Forest plot of the same top 30 MAL-associated proteins showing their association with emphysema progression. Points show $\hat{\beta}$ (log₂ RFU per HU progression) with 95% CI, from the model: $\log_2(\text{protein}) \sim \text{Change.lung.density.vnb.P2.P3} + \text{lung.density.vnb.P2} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{BMI} + \text{scanner} + \text{PC1-PC5}$. Vertical dashed line at zero.

- **Supp Figure C: Adaptive LASSO cross-validation curve.** Mean squared error from 10-fold CV plotted against $\log(\lambda)$ for the MAL-targeting adaptive LASSO. Shaded band: ± 1 SE. Dashed red line: selected $\lambda_{\min}$ and the number of proteins retained at that penalty.



Supplementary Figure C. Cross-validation curve for the adaptive LASSO ($\text{proRS}_{\text{MAL}}$). MSE (10-fold CV) is plotted against $\log(\lambda)$; shaded band shows ± 1 SE. The dashed line marks $\lambda_{\min}$, the penalty that minimises CV-MSE and determines how many proteins are retained in the risk score.

- **Supp Figure D: Proteomic risk score vs. CT MAL score.** Scatter of the MAL-targeting proteomic risk score (SD units, x-axis) against CT-derived meanmal (y-axis) for all $n = 1,306$ subjects. Fitted regression line with 95% CI band shown. $R^2 = 0.543$ confirms that the blood protein signature captures a meaningful fraction of CT-measured MAL variance.



Supplementary Figure D. Concordance between the MAL-targeting proteomic risk score (MAL_PRS_scaled , derived from 572 blood proteins) and the CT-derived MAL score (meanmal). Each point is one subject ($n = 1,306$); regression line with 95% CI shown. $R^2 = 0.543$.